

MOLLY XIE

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Education

University of Waterloo

Bachelor of Mathematics, Data Science (Co-op) | Avg: 86.3/100

ON, Canada

Relevant Coursework: Algorithms & Data Abstraction, Functional Programming, Calculus I–II, Linear Algebra

Actuarial Exam: SOA Exam P – Probability (Passed, May 2026)

Experience

Research Mentee, Quantitative Finance Project

May 2026 – Aug 2026

Women in Math Directed Reading/Research Program

University of Waterloo

- Conduct quantitative analysis of entry and exit strategies using historical stock and ETF data from Yahoo Finance API.
- Develop Python backtesting pipelines with pandas, NumPy, and matplotlib for SMA, EMA, RSI, and MACD analysis.
- Evaluate strategy robustness through parameter tuning and visualization against baseline investment strategies.
- Identify time-series trends and anomalies, translating analysis results into clear data-driven insights.

Administrative Assistant

Dec 2024 – Jan 2025

Hunan Chiyang Information Technology Co., Ltd.

Hunan, China

- Screened 50+ resumes and maintained applicant tracking system, reducing processing time for HR team.
- Maintained and updated applicant tracking spreadsheets in Excel, improving accuracy and consistency of records.
- Supported day-to-day administrative and HR-related tasks in a fast-paced office environment.
- Ensured data accuracy across internal records and inventory-related documentation.

Teaching Assistant

Jul 2024 – Sep 2024

New Channel International Education Group

Hunan, China

- Instructed 30+ students in IELTS and A-Level STEM subjects across 6 daily classes, with 80% achieving target scores.
- Prepared written feedback and progress reports, summarizing student performance and actionable recommendations.
- Analyzed learning patterns to identify gaps and adjust teaching strategies for different student needs.
- Collaborated with instructors to maintain consistent teaching quality in a fast-paced educational setting.

Projects

MatchaMatch | React, Vite, Tailwind CSS, Express, Firebase

Apr 2026

- Built a swipe-based platform for students to discover mentors, peers, and career opportunities through matching.
- Implemented real-time matching, messaging, and community board features using Firebase Firestore and Authentication.
- Designed a responsive Matcha-themed interface with React, Tailwind CSS, and Framer Motion for a polished experience.
- Collaborated on Firestore structure, security rules, and demo seeding workflows to support realistic user interactions.

SpatialVCS | Python, FastAPI, WebSocket, YOLOv8, LLM

Feb 2026

- Best Beginner Hack Winner, CTRL+HACK+DEL 2.0 2026
- Built a “Git for Reality” prototype that transforms physical spaces into searchable snapshots using object metadata.
- Designed a “Physical Diff” method to track object movement across commits with centroid-based distance rules.
- Enabled privacy-preserving natural language queries by storing embeddings and metadata rather than video frames.

SleepMatch | Python, Data Analysis, Streamlit

Dec 2025 – Jan 2026

- Built a sleep recommendation tool estimating optimal sleep duration based on age, activity level, and caffeine intake.
- Researched sleep guidelines from the NSF, CDC, and Harvard Medical School to inform recommendation logic.
- Transformed a command-line program into a deployed Streamlit web app with modular functions and input validation.
- Applied statistical reasoning to model relationships between lifestyle variables and recommended sleep duration.

Honors & Awards

University of Waterloo President’s Scholarship of Distinction

Sep 2025

Entrance scholarship awarded to students admitted with an admission average of 95% or above.

Technical Skills

Languages: Python, C, SQL

Data & ML: pandas, NumPy, matplotlib, scikit-learn, EDA, Data Visualization, Machine Learning Basics

Web & Development: React, Vite, Tailwind CSS, Express, Firebase, Streamlit

Tools & Technologies: Git, GitHub, Linux, VS Code, Excel